Fady Youssef
01/19/2026
CPSC 326, Spring 2026

Lean Structures Exercises

---

1) First three exercises for structures
   a) structure RectangularPrism where
      height : Float
      width : Float
      depth : Float

   b) def volume (p : RectangularPrism) : Float :=
       p.height * p.width * p.depth

   c) structure Point where
      x : Float
      y : Float

      structure Segment where
      p1 : Point
      p2 : Point

      def length (s : Segment) : Float :=
       Float.sqrt ((s.p2.x - s.p1.x)^2.0 +
                   (s.p2.y - s.p1.y)^2.0)

2) What are the four definitions that I have been using to describe lean.
   - Lean is a **Strict Pure Functional** Programming language with **Dependent types**.

3) Write the definition for a point in 3 space called Point3D that has 3 Float components: x, y, and z.

   structure Point3D where
   x : Float
   y : Float
   z : Float

4) Write a function that returns the smallest value of the three components of a Point3D called minimumComponent.

```
def minimumComponent (p : Point3D) : Float :=
if (p.x < p.y) && (p.x < p.z) then
  p.x
else if p.y < p.z then -- since in the first check we know y > x
  p.y
else
  p.z -- we now know z < x,y
```

5) Write a function that takes two Point3Ds and returns the midpoint called midpoint.

```
def midpoint (p1 : Point3D) (p2 : Point3D) : Point3D :=
{ x := (p1.x + p2.x) / 2.0,
  y := (p1.y + p2.y) / 2.0,
  z := (p1.z + p2.z) / 2.0 }
```